

Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The main goal of the project is to identify spending patterns, customer segments, product preferences, subscription behavior, and other factors that can help support business decisions.

2. Dataset Description

The dataset contains 3,900 purchase records and 18 columns.

The main information included in the dataset can be divided into three groups.

Customer information includes Age, Gender, Location, and Subscription Status.

Purchase information includes Item Purchased, Category, Purchase Amount, Season, Size, and Color.

Shopping behavior information includes Discount Applied, Promo Code Used, Previous Purchases, Payment Method, Frequency of Purchases, Review Rating, and Shipping Type.

There were 37 missing values in the Review Rating column.

3. Data Preparation and Cleaning Using Python

The dataset was first loaded and explored using Python and the pandas library. The structure of the dataset was checked using `df.info()`, while descriptive statistics were examined using `df.describe()`.

Missing values were checked throughout the dataset. The missing values in the Review Rating column were handled by replacing them with the median rating of the corresponding product category.

The column names were standardized using snake case to make them easier to use during the analysis and database operations.

Feature engineering was also performed during the preparation stage. An `age_group` column was created by grouping customers according to their age. A `purchase_frequency_days` column was also created by converting the purchase frequency categories into approximate numbers of days.

The consistency of the `discount_applied` and `promo_code_used` columns was also checked. Since the two columns contained redundant information, `promo_code_used` was removed.

After completing the cleaning and preparation process, the cleaned DataFrame was connected to a PostgreSQL database and loaded into the database for further analysis using SQL.

4. Data Analysis Using SQL

SQL was used to perform several business-oriented analyses and answer important questions related to customer behavior.

4.1 Revenue by Gender

The total revenue and average purchase amount were compared between male and female customers.

The analysis showed that the total revenue generated by male customers was higher because they represent a larger number of purchase records in the dataset. However, the average purchase amount between the two genders was relatively close.

4.2 High-Spending Discount Users

Customers who used discounts but still made purchases above the overall average purchase amount were identified.

This analysis helps determine whether discounts are being used by customers who are still willing to make relatively high-value purchases. It can also help the business evaluate the effectiveness of its discount strategy.

4.3 Top Five Products by Rating

The average review rating was calculated for each product, and the five products with the highest average ratings were identified.

These products can be considered for future marketing campaigns because high customer ratings can indicate positive customer satisfaction.

4.4 Shipping Type Comparison

The average purchase amount for Standard and Express shipping was compared.

The analysis showed that purchases using Express shipping had a higher average purchase amount than purchases using Standard shipping.

This relationship can be useful when evaluating premium shipping options and their connection with customer spending.

4.5 Subscribers and Non-Subscribers

Customer spending was compared between subscribers and non-subscribers.

The analysis considered both the total revenue and the average purchase amount for each subscription group. Non-subscribers generated more total revenue because they represent the larger group of customers. However, the average purchase amounts between subscribers and non-subscribers were relatively similar.

4.6 Discount-Dependent Products

The percentage of discounted purchases was calculated for each product.

The five products with the highest percentage of discounted purchases were identified. This analysis helps determine which products depend more heavily on discounts and can support better decisions when creating future discount strategies.

4.7 Customer Segmentation

Customers were divided into three segments according to their previous purchase history.

New customers are customers with up to five previous purchases.

Returning customers are customers with more than five and up to ten previous purchases.

Loyal customers are customers with more than ten previous purchases.

This segmentation helps the business understand the differences between customers based on their purchase history and allows different marketing strategies to be applied to each group.

The Loyal segment represents the largest customer segment in the analyzed data and contributes a significant portion of the total revenue.

4.8 Top Three Products per Category

The most purchased products were identified for each product category.

This analysis provides a clearer view of which products are most popular within Clothing, Footwear, Accessories, and Outerwear. The results can be used to improve product positioning and promotional campaigns.

4.9 Repeat Buyers and Subscriptions

Customers with more than five previous purchases were analyzed to determine whether repeat buyers are more likely to have a subscription.

The subscription rate among repeat buyers was compared with the overall subscription rate. The results did not show a large difference between the two groups.

This suggests that purchase history alone may not be enough to explain subscription behavior.

4.10 Revenue by Age Group

Customer ages were grouped into different age groups, and the total revenue generated by each group was calculated.

The analysis showed differences in revenue contribution between the age groups. The Senior group generated the highest revenue contribution in the analyzed dataset, followed by the Adult and Middle_aged groups.

This information can help the business focus marketing activities on the customer groups that generate higher revenue.

5. Power BI Dashboard

An interactive dashboard was created using Power BI to present the main findings of the analysis.

The dashboard provides a visual representation of customer behavior and makes it easier to understand the relationships between customer characteristics, purchasing behavior, products, subscriptions, discounts, and revenue.

Using Power BI also makes it possible to explore the data interactively and compare different customer groups and business factors.

6. Key Findings

The dataset contains 3,900 purchase records and provides information about customer demographics, purchases, and shopping behavior.

Male customers generate higher total revenue because they account for a larger number of purchase records, while the average purchase amounts between genders are relatively close.

Express shipping is associated with a higher average purchase amount compared with Standard shipping.

Loyal customers represent an important customer segment and contribute a large portion of the total revenue.

The age groups contribute different amounts of revenue, with the Senior group generating the highest revenue contribution in the analyzed data.

Some products have higher average review ratings and can be considered for promotional campaigns.

The dependence on discounts differs between products, which means that discount strategies can be adjusted according to individual product behavior.

Repeat buyers do not show a significantly higher subscription rate compared with the overall customer base based on the analysis performed.

7. Business Recommendations

The first recommendation is to increase the attractiveness of the subscription program by providing exclusive benefits, special offers, or loyalty rewards to subscribers.

The second recommendation is to develop a customer loyalty program that rewards repeat buyers and encourages Returning customers to become Loyal customers.

The third recommendation is to review the current discount strategy. Products with a high dependence on discounts should be monitored carefully to make sure that discounts are increasing sales without unnecessarily reducing profit margins.

The fourth recommendation is to promote highly rated and popular products through marketing campaigns and product recommendation sections.

The fifth recommendation is to focus marketing activities on the age groups that generate higher revenue and to monitor the performance of campaigns targeted at these groups.

The sixth recommendation is to evaluate Express shipping as a premium service. Since Express purchases have a higher average purchase amount than Standard purchases, the business can test premium shipping offers and measure their effect on customer spending and sales.

8. Tools and Technologies Used

Python and pandas were used for data loading, exploration, cleaning, and feature engineering.

PostgreSQL and SQL were used for storing the cleaned data and performing business-oriented analysis.

Power BI was used to create the interactive dashboard and present the results visually.

The customer shopping behavior CSV file was used as the main source of the transactional data.

9. Conclusion

This project demonstrates a complete data analysis process starting with raw customer shopping data and continuing through data cleaning, feature engineering, database integration, SQL analysis, and Power BI visualization.

The analysis provided useful information about customer spending, product ratings, discounts, subscriptions, shipping methods, customer loyalty, and age groups.

The results can help businesses make better decisions regarding customer retention, subscription programs, discount strategies, product promotion, and targeted marketing.

Overall, the project shows how data analysis tools can be used together to transform raw customer data into useful business insights and support data-driven decision-making.